CARLOS HUNG

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OPEN SOURCE PROJECTS

Livekit-Prism - A dual-channel multimodal AI voice gateway built with *Go* alongside a *Python* LiveKit agent plugin, broadcasting real-time PostHog analytics, AI Observability metrics, ClickHouse telemetry, and dynamic Chart.js graphics via WebRTC data channels.

Livekit.Plugins.Posthog - Commit 92d1ca1 – A PostHog plugin for LiveKit Agents providing AI Observability, latency waterfalls, and dual-channel multimodal tool execution.

Find more on GitHub.

Additional Accomplishments

- **Beat *Claude Opus 4.5*** with *1107* cycles in the Anthropic Performance Challenge. The objective was to optimize a slow "kernel" implementation by reducing the number of simulated CPU "clock cycles".
- Resolved *SpotBugs* violations **PR #208** in *Maestro* - Netflix's Workflow Orchestrator.

RECENT EXPERIENCE *v9.4*

Backend Software Developer | Feb 2023 to Pres | Atlanta, GA (Remote)

PRIZEPICKS develops an entertainment app allowing users to easily play Daily Fantasy Sports.

PrizePicks Platform – Rails | Ruby | REST API | RSpec | Postgres | Go

- Develop a Search and Personalization *MMP* utilizing Typesense, *pgvector*, MiniML, Two-Tower, *Thompson Sampling*, and gRPC. Transition the codebase from **Rust**/Python to **Go**/Python for rapid production deployment (PrizePicks Hackathon Spring'26)
- Develop *AuraPicks*, a ML engine that predicts the outcomes of future sports projections using an H2O.ai ML model trained on historical game data (PrizePicks Hackathon Summer'25)
- Improve and expand the REST API to build our newly centralized Back Office management system.
- Integrate with our simulation engine to provide the ability to automatically create and update projections (player cards) in real-time depending on the odds forecast before and during the game.

Full-Stack Software Developer | Sep 2021 to Feb 2023 | Coral Gables (Miami suburb), FL (Remote)

KIPU HEALTH develops EMR software solutions for behavioral health treatment centers.

Kipu EMR – **Rails** | Ruby | RSpec | JS/jQuery/Stimulus | Postgres | Sidekiq | AWS Cloud9, S3

- Dive deep into the medication taper algorithm which gradually decreases the total daily dose of medication depending on the frequency that must be administered. The entire UI is built with ERB.
- Sanitize the entire source code from cross-site scripting (XSS) vulnerability attacks (stored and reflected XSS) including applying OWASP HTML entity encoding prevention rules to wrap untrusted data.

Backend Software Developer | Feb 2020 to Sep 2021 | San Jose, CA (Remote)

SYNOPSYS (acquires *WhiteHat Security* from *NTT*) provides software solutions to empower developers to build secure web applications.

WhiteHat Security - ScanX – **Go** | JavaScript | Node.js | OWASP | Docker | Jenkins | REST | CI/CD (Azure)

- Provide solutions for easy and seamless adaptation of our ScanX in developers' development and testing (CI/CD) environments like Jenkins and Selenium, among others.
- Author and main contributor of the open-source Node.js package: newman-reporter-har, a custom Postman/Newman reporter that outputs a HAR file. A crucial solution to access Postman collections for vulnerability detection by ScanX, one of the Postman API Hack winners on Feb 16, 2021.

Full-Stack Contract Developer | Jun 2019 to Dec 2019 | San Francisco, CA (Hybrid)

WW, Weight Watchers rebranded, promotes wellness that works, inspiring healthy habits for real life.

Kurbo Wellness Platform – Rails | Ruby | RSpec | Postgres | **React** | HAML | Redis | Sidekiq | Git

- Design and develop a coach matching algorithm, to search the database for the best active coach, matching user's requested preferences based on system tags. Built with react-rails.
- Develop new action events to track user interactions. Integrate with registered business partners via REST APIs. Modify alert subsystem to CRUD threshold alerts, e.g.: height/weight or BMI changes.

CONDENSE EXPERIENCE

Senior Software Engineer / Lead / Architect / Front-End / Backend | 1984 to 2019 | SF Bay Area

Numerous companies from Startups to Mid-Size Businesses.

Highlight of Major Accomplishments – Python | Java | C/C++ | PostgreSQL | JS | HTML | GCP/AWS

- Search over 30M records using Lucene Solr 3.3. Provide full-text, faceted, and geospatial search.
- Architect, design, develop, and test a multithreaded Short Message Peer to Peer Protocol (SMPP) service using Netty to send 1M SMS messages per hour, 28 times faster than previous versions.
- Develop authorization model (backend) using Apache Shiro, a Java security framework, including permission configuration and user restriction.

SKILLS & EXPERTISE, AND MORE

- Skills – *Languages* (Golang, Python, Ruby, JavaScript/TypeScript, Java, PHP, Rust, C/C++) | *DB* (PostgreSQL, MySQL) | *NoSql* (Redis, Couchbase, Memcached) | *MVC* (Rails, Angular, React, Backbone.js, Drupal, Pyramid, Zend) | *REST* (Dropwizard) | *ORM* (SQLAlchemy, Doctrine) | *Search* (Solr) | *GUI* (React, HTML5, Bootstrap, SCSS) | *JS* (Lodash.js, Moment.js) | *AWS* (Cloud9, S3, EC2)
- *Education* – MS Computer Eng, Santa Clara Univ | BS Computer Science, Oregon State Univ
- *Foreign Language* - Se habla Castellano. Read, speak and write fluently Spanish from Venezuela, S.A.